

I started making this project as early as possible since I wanted to work on this alone. There were a lot of things that needed to be reconsidered before deciding to work on this project alone. But all those reasons flew out of the window when I thought to myself I would only be learning half or fourth and not fully if I continued on the groupings and especially, there is still a matter of dealing with discomfort because of my assigned group. And, I don't think I'll be able to work properly because of such discomfort. Fortunately, Mrs. Ortuyo made things possible for me to work on this project flexibly.

One of the mistakes I made while creating this project was starting immediately on the simulator instead of making the visualizer first – the NFA diagram for the binary clock. I realized that the structure of the diagram is the very logic of how the simulator works. Anyhow, I studied how the binary clock works. At first, I expected the input string to be only four (4) bits, so I designed the UI of the simulator in Figma with that in mind. But then, I took a step back if this was right, because I do not think the simulator would be this easy and simple – it's a project after all. I took my time understanding binary clocks and there were two types: 12-Hour format and 24-Hour format. I first thought it would be nice to have both options but creating two diagrams seems a hassle, so I simply opted for the 24-Hour format. I changed the designs in the Figma again before starting off on the HTML and CSS. Designing the UI was enjoyable but I stopped halfway and slowly started questioning what I was doing. Was I really doing this simulator correctly? Is this what Mrs. Ortuyo expects us to create the simulator? I had a lot of doubts despite my fast progress on making the simulator. So, I stopped halfway and inquired Mrs. Ortuyo of my current progress but I still haven't received any responses. So, I was in turmoil for two days overthinking what I was making. I decided to continue even if I feel like what I was making was wrong.

Alas, the CSS and HTML were ready. All I needed to do was the logic of the simulation. Instead of making a function for the entire state machine I decided to divide it into three parts which are state machines for the hours, minutes, and seconds. I started on the logic of distributing the index range for the three parts before implementing what were the restraints for each part. The hour should only accept four or five bits while the minutes and seconds strictly accept six (6) bits instead of seven (7) bits. My problem with the logic was that it was not in harmony with the logic of my NFA diagram when it comes to how the simulator stops once it encounters an invalid string or invalid input. When I made the NFA diagram through the use of JFLAP, it was truly amusing how easy it was to construct a finite automata there. I started off manually processing the inputs before simplifying the diagram. There were two parts that I considered a struggle when I made the diagram. First was the space character, I thought the lambda symbol which means no input also includes the space character, I later realized that it's not like that. I'll really have to input one space and that was enough. The second struggle was finding the construction of my diagram smooth and simple. This actually made me nervously laugh because what if I made this diagram all wrong? Slight mistakes, okay. But the possibility of constructing this diagram so incorrectly? That would be a nightmare.

The entire process of making this project was a back and forth type of double-checking. I kept on checking my HTML, CSS, and JavaScript and then the visualizer if it was in harmony until I thought to myself, *"Oh, this is good already!"*.